INCH-POUND

MIL-B-14837B(AR)
AMENDMENT 2
17 June 1991
SUPERSEDING
AMENDMENT 1
31 December 1975

MILITARY SPECIFICATION

BODY ASSEMBLY: 8619920

This amendment forms a part of Military Specification MIL-B-14837B(MU), dated 24 March 1972, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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1.1, Delete and substitute:

"1.1 Scope. This specification covers a modular unit of the Tank Commander's M36 and M36E1 Periscope, herein referred to as "body assembly".

2.1, Drawings, U.S. Army Frankford Arsenal

Delete: "F8298789" and "F8577787" together with their respective titles.

Delete: "Maximum" and substitute "Minimum" in title for D11729723.

Delete: "Minimum" and substitute "Maximum" in title for D11732101.

Add: "F10554769 10 inch Collimator (8 Degree Field of View)" and "F11746731 Test Stand".

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* 3.6, Delete and substitute:

"3.6 Sealing.- The body shall show no evidence of leakage in excess of 0.1 pound per square inch gage (psig) when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 1 hour. The interior atmosphere of the accepted body assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig)."

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3.7.1.10, Delete "8.0 seconds of arc" and substitute "10 seconds of arc"

4.5.2.3.2, Delete and substitute"

"4.5.2.3.2 Performance. Special testing equipment to test performance of the body assembly shall conform to the requirements of Drawing F8575858, Fixture, Body Assembly, Periscope: M36 and M36E1: Drawing 10540489, Set-up Instructions for Test Fixture F8575858, Periscope Body Assembly, M36 and M36E1; Drawing F10554769, 10 inch Collimator (8 Degree Field of View) and Drawing F11746731, Test Stand."

4.6.1, Delete and substitute:

"4.6.1 Storage temperature. This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The body assembly shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The body assembly shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual, tactile and audible examination of 4.6.4. The body assembly shall conform to the requirements of 3.4.1.

NOTE:The rate of temperature change in the test chamber shall not exceed 20 degree per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2."

4.6.2, Delete and substitute:

"4.6.2 Operating temperatures. This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The body assembly shall remain at this temperature for a minimum

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of five hours. At the completion of this five hour period, and while at -40°F, the body assembly shall be subjected to the tests in 4.6.7.13.1 to 4.6.7.13.3 inclusive, to determine compliance with the applicable requirements of 3.7.1.11.1 to 3.7.1.11.3 inclusive, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of five hours. At the completion of this five hour period and while at +150°F, the body assembly shall again be subjected to the test in 4.6.7.13.1 to 4.6.7.13.3 inclusive, to determine compliance with the applicable requirements of 3.7.1.11.1 to 3.7.1.11.3 inclusive, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room ambient temperature (+60° to +90°F) and the body assembly removed from the test chamber. Body assemblies which have passed the above tests shall be subjected to the test in 4.6.3."

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* 4.6.6, line 2, Delete "0.25 to 0.50" and substitute "0.0 to 0.50"

4.6.7, Delete and substitute:

"4.6.7 Performance. The following tests shall be performed with the body assembly mounted on special testing equipment conforming to 4.5.2.3.2. Mounting and securing of the body assembly on the test equipment shall be performed in accordance with Drawing A10540490, Operating Instructions for Test Fixture F8575858, Periscope Body Assembly, M36 & M36E1. During these tests, the 10" Collimator (8° Field of View) shall be adjusted for 1200 meters and the filter assembly shall be removed from the collimator eyepiece. Unless otherwise specified, the diopter adjustment knob of the body assembly eyepiece shall be set for best focus of the body assembly reticle and the target image."

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4.6.7.16.1, Delete and substitute:

"4.6.7.16.1 Interchangeability, infrared section.- This test shall be performed utilizing the minimum and maximum interchangeability gages in Drawings D11729723 and D11732101 respectively, to determine compliance with 3.2.1b."

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6.1, Delete and substitute:

6.1, Intended use. The body assembly: 8619920, is a modular unit of the Tank Commander's Periscope: M36 and M36E1, primarily used in the fire control system of the 105MM, Gun Tank: M60 Series."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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